

Group 3 — Dictionary learning (Sparse Autoencoders)

Where this sits. This is the inverse half of the superposition story. Group 2 (the *Landscape* survey, Part II) asks the *forward* question — given a set of sparse features and a budget of $d = \dim V$ dimensions, what packing does training choose? Group 3 asks the *inverse*: handed only the activations a trained model actually produces, can we **recover** the overcomplete feature frame that generated them? The mathematical genre is **overcomplete dictionary learning** / **sparse coding**; the recurring anxiety, stated by the authors themselves, is that there is **no ground-truth objective** for what a "feature" is, so the recovery is optimised against a proxy. All four papers are from the Anthropic / Transformer Circuits lineage. Conventions (feature = direction; dictionary map $D : \mathbb{R}^F \rightarrow V$; encoder/decoder; $GL(V)$ gauge; adjoint W^* vs. dual $W^\vee$) are those of the *ML Concepts* primer.

The shared question

Fix an activation site inside a transformer — an MLP's neuron vector, or (later) the residual stream $V = \mathbb{R}^d$. Feed the model a large corpus; collect the cloud of activation vectors $\{x^j\} \subset V$ it produces. The **Linear Representation Hypothesis** (the *Landscape* survey, Part I.1) asserts that each such vector is a *sparse* linear combination of a fixed family of **feature directions** $\{d_i\}_{i=1}^F$,

$$x^j \approx b + \sum_{i=1}^F f_i(x^j) d_i, \quad f_i(x^j) \geq 0, \quad \#\{i : f_i \neq 0\} \text{ small},$$

with $F \gg d$ (overcomplete) because superposition lets the model traffic in far more features than dimensions (Part I.2). Collecting the directions into a **dictionary map** $D : \mathbb{R}^F \rightarrow V$, $De_i = d_i$, this is $x \approx b + Df(x)$: a linear matrix factorisation in which D is a fat, non-injective matrix and $f(x)$ is a sparse code.

The **shared question** of this group is the *inverse problem* this poses. The model never hands you D or f ; it hands you only the vectors x^j . Recovering the pair (D, f) from $\{x^j\}$ alone — with D overcomplete and f required sparse — is exactly classical **sparse dictionary learning** (Olshausen–Field, k -SVD). Two strategies for getting interpretable units divide the group:

1. **Post-hoc recovery.** Leave the trained model untouched and *learn* D from its activations with a sparse autoencoder (SAE) — the dominant line: Bricken et al. [3], Templeton et al. [7], and (across layers/models) Lindsey et al. [6].
2. **Architectural alignment.** Change the architecture so that features align to the privileged neuron basis *in the first place*, making the dictionary trivially the identity — Elhage et al. [2].

They belong together because they answer the same recovery question under the same superposition hypothesis, share the same Lagrangian (reconstruction + sparsity), and lean on the same near-orthogonality of the recovered frame. They diverge on *when* you intervene (after training vs. during) and on *what* the dictionary spans (one site vs. many layers vs. two models).

The papers

Towards Monosemanticity (the existence proof)

Bricken et al. [3] is the foundational demonstration. The target is deliberately minimal: a one-layer transformer with a 512-neuron ReLU MLP — "the simplest language model we profoundly don't understand," because the MLP nonlinearity is exactly what Elhage et al. [1]’s exact-rewrite program (Group 1) cannot freeze. They decompose the MLP activation $x \in V = \mathbb{R}^{512}$ as in (1) and fit it with a **sparse autoencoder**: a wide, shallow autoencoder whose hidden layer is the sparse code. Writing the encoder/decoder explicitly,

$$f(x) = \text{ReLU}(W_e(x - b_d) + b_e), \quad \hat{x} = b_d + W_d f(x),$$

with the columns of W_d the (unit-norm) feature directions d_i and the input/output biases *tied* (subtract a fixed b_d , reconstruct it back). It is trained by Adam to minimise reconstruction error plus an ℓ_1 sparsity penalty:

$$\mathcal{L} = \mathbb{E}_x \left[\|x - \hat{x}\|_2^2 + \lambda \sum_i f_i(x) \|d_i\| \right].$$

The autoencoder is **overcomplete**: F ranges from $1 \times (512 \text{ features})$ to $256 \times (131,072)$, with detailed analysis of the run "A/1" at $F = 4096$. Two empirical regularities are named here and recur throughout the group. **Dead features**: some hidden units cease to fire on any input, wasting dictionary capacity; they are revived by *resampling* (reinitialising a dead unit toward a poorly-reconstructed datapoint). **Feature splitting**: as F grows, one coarse feature resolves into several finer ones (a single base64 feature splits into three), so there is *no canonical feature count* — the dictionary width is a resolution knob, not a fact about the model.

Two arguments in this paper are load-bearing for the whole group. First, the **recovery is non-trivial**: computing $f(x)$ from x given D asks to invert a fat ($d \times F$, $F \gg d$) matrix — impossible *except* that f is sparse, which is precisely **compressed sensing** (NP-hard exactly; here approximated by a single amortised ReLU encoder rather than an iterative solver, deliberately, so the SAE is "no stronger" at recovery than the MLP that produced the activations). Second, a sharp argument against the architectural route (taken up by SoLU): under **cross-entropy** loss a model often achieves lower loss with a *polysemantic* neuron than a monosemantic one, so merely forcing activation sparsity does not buy monosemanticity. Tellingly, the SAE itself is fit under **MSE**, under which the monosemantic and polysemantic optima can coincide — which is why sparsity *can* discourage superposition inside the learned dictionary even though it cannot in the host model. The paper closes on the group’s defining unease: "are models in some fundamental sense composed of features, or are features just a convenient post-hoc description?" — left explicitly open.

Scaling Monosemanticity (the same recipe at frontier scale)

Templeton et al. [7] carries the identical recipe to Claude 3 Sonnet, a production model, and to the **residual stream** (middle layer) rather than an MLP. Activations are first scalar-normalised so their mean squared ℓ_2 norm equals $D = \dim V$; then $x \in \mathbb{R}^D$ is decomposed as

$$\hat{x} = b^{dec} + \sum_{i=1}^F f_i(x) W_{:,i}^{dec}, \quad f_i(x) = \text{ReLU}(W_{i,\cdot}^{enc} \cdot x + b_i^{enc}),$$

trained against the **decoder-norm-weighted** ℓ_1 objective

$$\mathcal{L} = \mathbb{E}_x \left[\|x - \hat{x}\|_2^2 + \lambda \sum_i f_i(x) \|W_{:,i}^{dec}\|_2 \right].$$

The factor $\|W_{:,i}^{dec}\|_2$ is not cosmetic: it lets the unit-normalised columns be read as feature *directions* and the products $f_i\|W_{:,i}^{dec}\|_2$ as feature *activations*, and it blocks the degenerate "cheat" of shrinking f_i while inflating $\|d_i\|$ to evade the penalty at fixed reconstruction. Three SAEs were trained at $F \approx 1\text{M}, 4\text{M}, 34\text{M}$ features ($\lambda = 5$). Because the loss is "a standard ML problem," they fit **scaling laws** (compute $\propto F \times \text{\#steps}$) to choose the 34M run's training length — treating dictionary recovery as an ordinary optimisation target.

The mathematically suggestive result is the **concept-frequency law**. For families of concepts (elements, cities, animals, foods) the probability that the dictionary contains a dedicated feature for a concept rises with that concept's frequency in the training data; rescaling the frequency axis by the number of *alive* features collapses the three runs onto one curve "resembling a **sigmoid in log-frequency space**," with the 50%-inclusion threshold near the inverse of the alive-feature count and a conjectured tie to **Zipf's law** (the millionth feature represents a concept $\sim 10\times$ rarer than the hundred-thousandth). **Dead features** scale adversely: $\approx 2\%, 35\%, 65\%$ dead at 1M/4M/34M — the bigger the dictionary, the more capacity wasted. The discussion is candid about the limits: an **"Inability to Evaluate"** (no principled objective; the MSE/ ℓ_1 trade-off is admitted unprincipled), and **cross-layer superposition** — features "smeared" across layers that no single-site SAE can cleanly isolate, motivating crosscoders.

Softmax Linear Units (the architectural alternative)

Elhage et al. [2] takes the other fork: rather than *recover* a dictionary post-hoc, *redesign* the MLP so features align to neurons during training, making the privileged neuron basis (the *ML Concepts* primer §1.5) the dictionary. The device is the **softmax linear unit**, a drop-in replacement for the GeLU activation (recall $\text{GeLU}(x) \approx \text{sigmoid}(1.7x) \odot x$; replace the binary sigmoid by its multivariate sibling softmax):

$$\text{SoLU}(x) = x \odot \text{softmax}(x).$$

Acting on a whole neuron vector, softmax implements **lateral inhibition**: a large coordinate suppresses its neighbours, e.g. $\text{SoLU}(4, 1, 4, 1) \approx (2, 0, 2, 0)$, encouraging few-hot activations. The deeper rationale is **superlinearity**: like $\text{ReLU}(x)^2$ or $\exp(x)$, SoLU satisfies $f(x/\sqrt{N}) < f(x)/\sqrt{N}$, so *spreading* a feature across N neurons shrinks it — a built-in penalty on superposition that favours basis-aligned features. Empirically this raises the fraction of "interpretable" neurons from $\sim 35\%$ to $\sim 60\%$ at no measured loss cost.

The mathematics of the catch — the **"no free lunch"** — is the most interesting part for this survey. Bare SoLU costs performance (as superposition predicts it must), so a **LayerNorm** is appended: $f(x) = \text{LN}(\text{SoLU}(x)) = \text{LN}(x \odot \text{softmax}(x))$. But LayerNorm is invariant to input scaling, $\text{LN}(\alpha x) = \text{LN}(x)$, so the softmax *denominator cancels*:

$$\text{LN}(\text{SoLU}(x)) = \text{LN}\left(x \frac{\exp(x)}{\sum_i \exp(x_i)}\right) = \text{LN}(x \odot \exp(x)).$$

A feature that SoLU suppressed by spreading it thin is rescaled back up by the normaliser — superposition is **smuggled through** in small activations and recovered downstream. So the architecture makes some features visible by hiding others more deeply. The authors read this as *evidence for* the superposition hypothesis: you cannot cheaply legislate it away.

Sparse Crosscoders (one dictionary across layers and models)

Lindsey et al. [6] generalises the SAE from a single site to several at once. A **crosscoder** reads a set of layers L and reconstructs each. Its code sums encoder contributions across layers,

$$f(x_j) = \text{ReLU}\left(\sum_{l \in L} W_{enc}^l a^l(x_j) + b_{enc}\right), \quad a''(x_j) = W_{dec}^l f(x_j) + b_{dec}^l,$$

and is trained with a reconstruction term *summed over layers* plus a sparsity penalty that is the signature object here, the ℓ_1 -**of-norms**:

$$\mathcal{L} = \sum_{l \in L} \|a^l(x_j) - a''(x_j)\|^2 + \sum_i f_i(x_j) \left(\sum_{l \in L} \|W_{dec,i}^l\| \right).$$

The penalty weights feature i by the *sum* over layers of its per-layer decoder-vector norms $\|W_{dec,i}^l\|$ (each itself an ℓ_2 norm) — an ℓ_1 across layers of ℓ_2 -within-layer. The deliberate contrast is with the "natural" choice, the ℓ_2 -of-norms $\sqrt{\sum_l \|W_{dec,i}^l\|^2}$ (treating all layers' decoder weights as one vector). The ℓ_1 -of-norms is chosen for two reasons: it makes the crosscoder loss *directly comparable* to a sum of per-layer SAE losses (no spurious "bonus" for spreading a feature across layers), and it does not reward spreading, so it surfaces a mix of **shared and site-specific** features — whereas ℓ_2 -of-norms recovers only shared ones.

Two applications follow. **Cross-layer features**: one dictionary resolves cross-layer superposition and tracks a feature persisting through the residual stream, collapsing the per-layer-SAE picture's "duplicate" features (a circuit of three features that per-layer SAEs render as thirteen). Notably the crosscoder defines a feature as "the same" across layers if it *activates on the same data*, allowing the decoder *direction* to drift across layers — and empirically it does. **Model diffing**: run one dictionary over the *same* site in *two* models (a base model and its finetune, or two independent models) to read off which features are shared and which are model-specific. This is the bridge to Group 4 (the *Landscape* survey): *transcoders* (predict a later activation, modelling *computation* not representation) and crosscoders are exactly what attribution graphs are built from.

Common mathematical core

Strip the engineering and all four reduce to one problem: **overcomplete dictionary learning with an ℓ_1 relaxation of ℓ_0 sparsity**. Three pillars hold it up.

(i) **The ℓ_1 surrogate for ℓ_0** . What we want is for $f(x)$ to have few nonzeros — minimal ℓ_0 "norm." That is combinatorial and NP-hard, so every paper substitutes the convex ℓ_1 penalty $\lambda \sum_i f_i \|d_i\|$, the same surrogate that makes compressed sensing tractable. The substitution is the source of the group's shared pathologies (below) and is the same "reconstruction + sparsity" Lagrangian $\mathbb{E}\|x - \hat{x}\|^2 + \lambda \rho(f)$ that runs through Groups 2, 4, and 6 (the *Landscape* survey, Part III.4).

(ii) **Reliance on near-orthogonality, $G \approx I$** . Recovery of a sparse code from a fat dictionary is guaranteed (compressed sensing) only when the dictionary is **incoherent** — its columns nearly orthogonal, i.e. the **Gram matrix** $G = D^*D$ has small off-diagonal, $G \approx I$. This is the same Johnson–Lindenstrauss / Thomson geometry Group 2 derives on the forward side (the *Landscape* survey, Part I.2, §4). The off-diagonal of G is interference; feature splitting is G revealing finer block structure as F grows; the "weight interference" central mode the monosemanticity papers see when comparing two models' features is G 's off-diagonal made visible.

(iii) **Inverse to Group 2’s forward problem, via one theorem read twice.** Group 2 uses compressed sensing to *upper-bound* how many features can be packed (a forward statement about the embedding W); Group 3 *is* the recovery algorithm that same theorem promises (an inverse statement, recovering f from $x \approx Df$). The SAE encoder is a learned, amortised stand-in for the iterative sparse solver — one feed-forward pass $\text{ReLU}(W_e x + b)$ instead of an optimisation per datapoint. This is the cleanest place in the whole corpus where two camps share a single piece of mathematics (the *Landscape* survey, Part III.5).

How they diverge

- **Post-hoc recovery vs. architectural change.** The three SAE/crosscoder papers leave the model fixed and learn D afterwards; SoLU rewrites the activation function so the dictionary is forced to be (near) the standard basis at training time. Recovery is non-invasive but yields a dictionary with no privileged status; the architectural route bakes interpretability in but — the LayerNorm no-free-lunch — cannot fully suppress what superposition wants.
- **Single-site vs. cross-layer vs. cross-model.** Towards/Scaling fit one dictionary to one activation site (an MLP, then the mid residual stream). Crosscoders fit *one* dictionary jointly across many layers (ℓ_1 -of-norms) and even across two models (diffing). The penalty geometry is exactly what changes: a per-site ℓ_1 becomes an ℓ_1 of per-layer ℓ_2 norms.
- **Toy vs. frontier scale.** Towards Monosemanticity is a 512-neuron one-layer testbed where logit effects are essentially linear and ground-truth-ish proxies exist; Scaling Monosemanticity is a proprietary frontier model with 34M features, scaling laws, and heavy hedging. Formality drops as scale rises (the *Landscape* survey, Part V).
- **What is modelled.** SAEs/crosscoders here model *representation* (a basis for activation space). The transcoder variant flagged for Group 4 models *computation* (a map between spaces) — a divergence in what "correct" means, taken up below and in IV.2 of the *Landscape* survey.

What they answer, and what they don’t

What they answer. Superposition is real and dictionaries can be recovered: SAEs extract thousands-to-millions of features that are far more monosemantic than neurons, causally steer behaviour, and recur across models (universality). The recipe scales (towards $\rightarrow$ scaling), generalises across layers and models (crosscoders), and even has a tidy quantitative law (the concept-frequency sigmoid). The architectural route shows the choice of nonlinearity *measurably* moves polysematicity.

What they don’t. The honest gaps are mostly mathematical, and the authors say so.

- **No ground-truth feature, no principled objective.** There is no definition of "feature" against which to score D ; the trained objective is reconstruction + sparsity, an admitted *proxy* for interpretability ("Inability to Evaluate"). **MSE-vs- ℓ_1 is unprincipled:** nobody knows the right trade-off λ , or how to know one was made well.
- **Shrinkage.** The ℓ_1 penalty biases active f_i downward (it penalises magnitude, not just support), so reconstructions systematically undershoot — the standard LASSO shrinkage, here a bias in every feature activation. (JumpReLU/threshold encoders are partial fixes.)
- **Dead features and no canonical width.** Up to 65% of a large dictionary can be dead; feature splitting means F is a resolution dial, not a count — there is no "right number of features."

- **Cross-layer superposition.** Features smeared across layers defeat any single-site SAE; cross-coders mitigate but do not solve it (the authors call it "very fundamental").
- **Gauge-dependence of the loss (the deep one).** The reconstruction loss $\|x - \hat{x}\|_2^2$ and every decoder-cosine comparison are **metric-dependent**: they silently insert the Euclidean $G = I$ as an inner product on V — yet Groups 2 and 4 insist that same residual stream has **no privileged basis**, a $GL(V)$ gauge symmetry (the *Landscape* survey, Parts I.4, III.2). Under a gauge $x \mapsto Mx$ the ℓ_2 loss is *not* invariant, so what the SAE deems a good reconstruction depends on a coordinate choice the architecture never supplied. The empirical escape hatch — Adam and training *do* break the symmetry (Elhage et al. [4]), so the data-induced metric is not arbitrary — is a crutch, not a principled choice of metric.
- **The framing question, unresolved:** "is the model *made of* features, or are features a post-hoc description?" Universality is suggestive but not decisive.

Connections

To Group 2 (the forward problem). Group 3 is the literal inverse of Group 2's packing problem: where toy models ask how an overcomplete frame is *stored*, SAEs ask how it is *read back out*. Both pivot on the same Gram matrix $G = D^*D$ and the same compressed-sensing theorem, read forward (capacity bound) and backward (recovery). Feature splitting (Group 3) and the polytope/tegm geometries (Group 2) are two views of the same G .

To Group 4 (attribution). The transcoder/crosscoder machinery here is the substrate of the circuit-tracing program: Ameisen, Lindsey, et al. [8] replaces each MLP with a *cross-layer transcoder* — a Group-3 sparse model that predicts *computation* — and reads off a linear attribution graph; Lindsey et al. [10] applies it at frontier scale. So Group 3 supplies the nodes that Group 4's edges connect. The shift from *representation* to *computation* is also where mechanistic faithfulness becomes a worry (Anthropic Interpretability Team [9]): a transcoder can be sparse, monosemantic, low-error, and still implement the wrong algorithm.

To the gauge/metric blind spot. The pointed open question (the *Landscape* survey, Part VI.1): **if the ℓ_2 reconstruction loss is gauge-dependent, what *should* an SAE minimise?** A genuinely basis-free objective would minimise reconstruction error in a *derived* metric, not the default Euclidean one. Park et al. [5] answers the analogous question for the model's *output* space with the **causal inner product** $g_C = \text{Cov}(\gamma)^{-1}$ (a whitening/Mahalanobis form); the corresponding metric for the residual-stream *interior* that an SAE acts on is, as far as this corpus shows, unwritten. Recasting interference and feature interaction in metric-free terms — image-in-kernel / virtual weights rather than the $G = I$ Gram matrix (the *ML Concepts* primer §1.3) — is one concrete shape the fix could take.

For the mathematician

Three threads are worth pulling.

Recovery guarantees vs. the amortised encoder. Classical **compressed sensing** gives clean theorems: a k -sparse $f \in \mathbb{R}^F$ is exactly recoverable from $x = Df$ by ℓ_1 minimisation provided D satisfies a restricted-isometry / incoherence condition (coherence $\mu(D) = \max_{i \neq j} |\langle d_i, d_j \rangle|$ small, i.e. $G \approx I$), with sample complexity $\dim V \gtrsim k \log(F/k)$. The SAE does *not* run that minimisation: it replaces the per-datapoint convex program by a single **learned, amortised** affine-then-ReLU

encoder $f(x) = \text{ReLU}(W_e x + b)$. The open mathematical questions are sharp. (a) When does an amortised one-layer encoder *realise* the ℓ_1 recovery map — for what D is the optimal sparse code a pointwise ReLU of a linear readout? (b) The encoder W_e is essentially asking for an approximate left-inverse of the fat D ; the natural object is the **pseudoinverse** $D^+ = (D^* D)^{-1} D^*$ premised on G invertible — but in superposition G is singular ($F > d$ forces $\ker D \neq 0$), so what is recovered is governed by the conditioning of G restricted to the active support. (c) The known ℓ_1 **shrinkage** bias has a closed form for LASSO; quantifying the analogous bias of the amortised encoder, and whether a thresholded (JumpReLU) encoder removes it, is concrete.

The missing principled objective. The field optimises $\mathbb{E}\|x - \hat{x}\|_2^2 + \lambda\rho(f)$ and openly calls it a proxy. A mathematician's instinct is to ask what *invariant* this should be a relaxation of. The ℓ_0 "true" objective is gauge-invariant (counting nonzeros of f ignores any metric on V), but the ℓ_2 reconstruction term is *not*. So the current objective mixes a gauge-invariant sparsity term with a gauge-variant fidelity term — and the trade-off λ between them is doubly ill-posed. Is there an objective whose minimiser is independent of the $GL(V)$ frame? Candidates: measure reconstruction error in a model-derived metric (a residual-stream analogue of $g_C = \text{Cov}(\gamma)^{-1}$), or replace $\|x - \hat{x}\|_2^2$ by a divergence on the *downstream behaviour* (KL on the next-token distribution), which is gauge-invariant because it is read through the model's own maps. Either reframes "good dictionary" as a property of the model's function, not of an arbitrary coordinate system.

The gauge-invariant reformulation question. Stated cleanly: the residual stream V carries a $GL(V)$ symmetry; the SAE's loss, its decoder cosine-similarities, and its Gram-matrix diagnostics all live in the metric-dependent column (the *Landscape* survey, Part III.1) and are therefore not canonically meaningful. Two genuinely invariant objects survive and could anchor a reformulation: the natural **dual pairing** (a feature read by a downstream covector, $\langle w, d_i \rangle$, needs no metric) and the **virtual weight / image-in-kernel** relation (whether one component's write subspace lands in another's kernel). A worthwhile programme: re-derive "feature," "interference," and "reconstruction quality" using only these, ask whether dictionary recovery can be posed entirely in the quotient/dual language, and so decide whether the gauge-dependence of the ℓ_2 loss is a harmless convention that training fixes or a genuine obstruction to a canonical notion of "the model's features." This is the leading candidate, in this group, for where the mathematics actually is.

References

- [1] N. Elhage, N. Nanda, C. Olsson, et al. *A Mathematical Framework for Transformer Circuits*. Transformer Circuits Thread. 2021. URL: <https://transformer-circuits.pub/2021/framework/index.html> (visited on 05/29/2026).
- [2] N. Elhage, T. Hume, C. Olsson, et al. *Softmax Linear Units*. Transformer Circuits Thread. 2022. URL: <https://transformer-circuits.pub/2022/solu/index.html> (visited on 05/29/2026).
- [3] T. Bricken, A. Templeton, J. Batson, et al. *Towards Monosemanticity: Decomposing Language Models With Dictionary Learning*. Transformer Circuits Thread. 2023. URL: <https://transformer-circuits.pub/2023/monosemantic-features/index.html> (visited on 05/29/2026).
- [4] N. Elhage et al. *Privileged Bases in the Transformer Residual Stream*. Transformer Circuits Thread. 2023. URL: <https://transformer-circuits.pub/2023/privileged-basis/index.html> (visited on 05/29/2026).

- [5] K. Park, Y. J. Choe, and V. Veitch. *The Linear Representation Hypothesis and the Geometry of Large Language Models*. arXiv:2311.03658; ICML 2024. 2023. URL: <https://arxiv.org/abs/2311.03658> (visited on 05/29/2026).
- [6] J. Lindsey, A. Templeton, J. Marcus, et al. *Sparse Crosscoders for Cross-Layer Features and Model Diffing*. Transformer Circuits Thread. 2024. URL: <https://transformer-circuits.pub/2024/crosscoders/index.html> (visited on 05/29/2026).
- [7] A. Templeton, T. Conerly, J. Marcus, et al. *Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet*. Transformer Circuits Thread. 2024. URL: <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html> (visited on 05/29/2026).
- [8] E. Ameisen, J. Lindsey, et al. *Circuit Tracing: Revealing Computational Graphs in Language Models*. Transformer Circuits Thread. 2025. URL: <https://transformer-circuits.pub/2025/attribution-graphs/methods.html> (visited on 05/29/2026).
- [9] Anthropic Interpretability Team. *A Toy Model of Mechanistic (Un)Faithfulness*. Transformer Circuits Thread. 2025. URL: <https://transformer-circuits.pub/2025/faithfulness-toy-model/index.html> (visited on 05/29/2026).
- [10] J. Lindsey, W. Gurnee, E. Ameisen, et al. *On the Biology of a Large Language Model*. Transformer Circuits Thread. 2025. URL: <https://transformer-circuits.pub/2025/attribution-graphs/biology.html> (visited on 05/29/2026).